



GP Demand and Demographic Change: Forecasting Disease Prevalence in Scotland to 2043

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Introduction

General Practice (GP) is the bedrock of the Scottish public health system, and demand is steadily increasing as Scotland's population continues to age. This study integrates GP registration and practice-level disease prevalence data[6, 5, 9] with 2022-based ONS population projections [8] to project future health care demand across eight of the most common diseases in Scotland: diabetes, depression, hypertension, asthma, cancer, coronary heart disease, chronic kidney disease, and stroke/TIA. I employ a Bayesian hierarchical model to quantify variation across practices and geographies, and the historically-based disaggregations of the ONS projections to forecast disease burden scenarios through 2043.

Preprocessing and Exploratory Data Analysis (EDA)

- **Data assembly.** Linked *GP Practice Population Demographics*, *GP Practice Contact Details*, *SIMD 2020v2*, and *Data Zone 2011*, and practice-level disease prevalence datasets from PHS with population projections by ONS at the council area level in 2018 and the national level in 2022. [5, 6, 4, 9, 1, 2].
- **Harmonization.** Reconciled age and sex totals, annualized quarterly data, constrained rates to $[0, 1]$, and engineered predictors: $\log(\text{patients/GPs})$, nearest practice distance, SIMD decile. Centered and scaled continuous variables.
- **Population projections.** Disaggregated 2022-based national projections by age and sex to council areas using 2018 projection-based council area proportions, further disaggregated to council areas using historical 2022-based data zone proportions, and reaggregated data zones into HSCPs.[8, 7, 3].

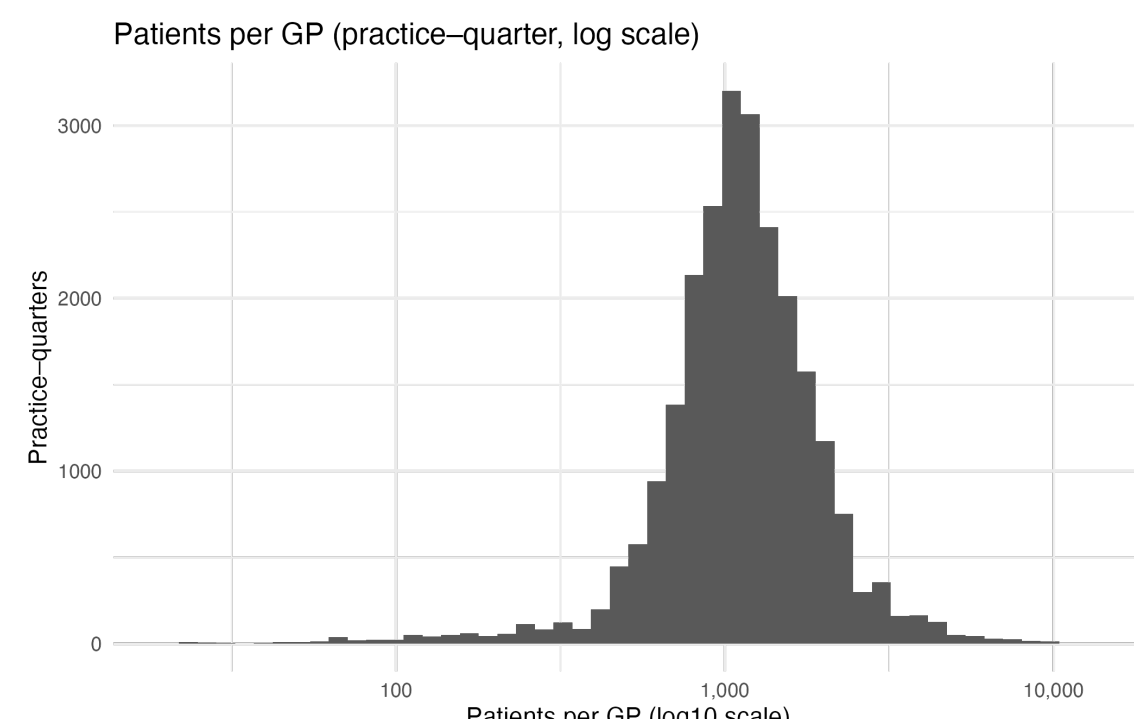


Figure 1. Distribution of patients per GP (log-scale).

- **Data visualisation.** Patients per GP is right-skewed, prevalence shows strong age-sex structure (including interaction), and deprivation/distance effects are disease-specific.

- **Covariate selection.** Retained age, sex, age:sex, $\log(\text{patients/GPs})$, and on a disease by disease basis SIMD decile and nearest practice distance.

Model Specification

- **Specification.** Bayesian GLMs were fitted using two distribution families: Binomial models (with disease counts as the response) and Beta models (with disease rates as the response). All models included random intercepts for practice, HSCP, and health board, together with fixed effects for age, sex, and $\log(\text{patients/GPs})$. Within each family, models with and without SIMD decile and nearest-practice distance as additional fixed effects were compared, and the specification that improved model fit in a sample of practices for each disease was retained. For the Beta family, a zero-inflated Beta (ZOIB) variant including all covariates was also evaluated.

- **Priors.** The following priors, selected to be weakly informative, were used in the models:

Binomial	Beta	ZOIB
$y_i \mid n_i, p_i \sim \text{Binomial}(n_i, p_i)$	$z_i \mid \mu_i, \phi \sim \text{Beta}(\mu_i \phi, (1 - \mu_i) \phi)$	$z_i \mid \pi_{0,i}, \mu_i, \phi, \pi_{1,i} \sim \text{ZOIB}(\pi_{0,i}, \mu_i, \phi, \pi_{1,i})$
$\text{logit}(p_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + u_{\text{prac}, j[i]} + u_{\text{HSCP}, k[i]} + u_{\text{HB}, m[i]}$	$\text{logit}(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + u_{\text{prac}, j[i]} + u_{\text{HSCP}, k[i]} + u_{\text{HB}, m[i]}$	$\text{logit}(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + u_{\text{prac}, j[i]} + u_{\text{HSCP}, k[i]} + u_{\text{HB}, m[i]}$
$\beta_j, \alpha \sim \mathcal{N}(0, 1)$	$\beta_j, \alpha \sim \mathcal{N}(0, 1)$	$\text{logit}(\pi_{0,i}) = \mathbf{z}_i^\top \boldsymbol{\gamma}^{(0)}$
$u_{\text{prac}}, u_{\text{HSCP}}, u_{\text{HB}} \sim \mathcal{N}(0, \sigma^2)$	$u_{\text{prac}}, u_{\text{HSCP}}, u_{\text{HB}} \sim \mathcal{N}(0, \sigma^2)$	$\text{logit}(\pi_{1,i}) = \gamma_0^{(1)}$
$\sigma_{\text{prac}}, \sigma_{\text{HSCP}}, \sigma_{\text{HB}} \sim t_3^+(0, 2.5)$	$\sigma_{\text{prac}}, \sigma_{\text{HSCP}}, \sigma_{\text{HB}} \sim t_3^+(0, 2.5)$	$\beta_j, \alpha, \gamma_j^{(0)}, \gamma_0^{(1)} \sim \mathcal{N}(0, 1)$
	$\phi \sim \text{Gamma}(2, 0.1)$	$u_{\text{prac}}, u_{\text{HSCP}}, u_{\text{HB}} \sim \mathcal{N}(0, \sigma^2)$
		$\sigma_{\text{prac}}, \sigma_{\text{HSCP}}, \sigma_{\text{HB}} \sim t_3^+(0, 2.5)$
		$\phi \sim \text{Gamma}(2, 0.1)$

Diagnostics

- **Within-family fit.** Final beta and binomial models were chosen from among the reduced (excluding SIMD/nearest practice distance), expanded (including SIMD/nearest practice distance), and, for beta, the ZOIB models. Selection was based on LOO performance (see Table 1) in a sample of 40 practices, after which the chosen models were run on the full practice-level dataset.

Disease	Beta			Binomial	
	Reduced	Expanded	ZOIB	Reduced	Expanded
Asthma	6504	6504	6498	-9228	-9228
Cancer	11209	11209	11202	-5798	-5798
Chronic Kidney Disease	12601	12600	12595	-4312	-4313
Coronary Heart Disease	11765	11768	11761	-5628	-5628
Depression	7656	7656	7649	-15866	-15866
Diabetes	9651	9651	9645	-7054	-7054
Hypertension	9079	9083	9075	-8249	-8249
Stroke/TIA	11780	11782	11776	-4790	-4791

Table 1. Leave-one-out (LOO) model comparison by disease. Values are expected log predictive density (ELPD), rounded to nearest integer. Best model within each family in bold.

- **Inter-family fit.** Model selection between the Beta and Binomial families was determined by posterior predictive checks. These revealed that the Binomial models were systematically underdispersed, whereas the Beta models better captured variability through their precision parameter. Consequently, the Beta model was selected for each disease.

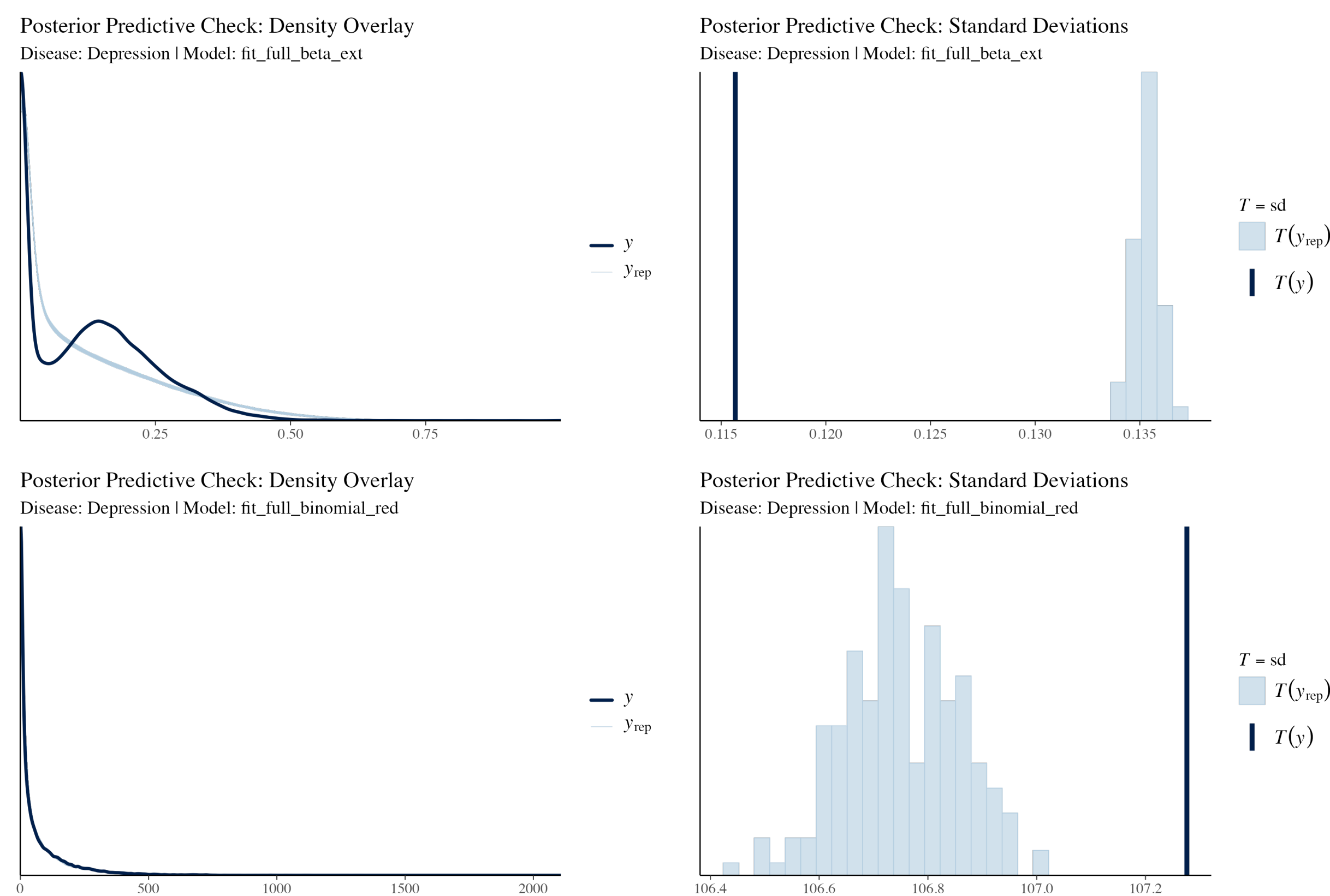


Figure 2. Posterior predictive checks for depression. The Binomial models were underdispersed and the Beta model is overdispersed, but greater than necessary variability was seen as preferable to less.

- **Convergence.** Gelman–Rubin diagnostics ($\hat{R} \leq 1.01$) and trace plots across all models in both families, indicating good convergence.
- **Projection.** The models selected, the disaggregated 2022-based population projections by age and sex were then used with the Beta models, holding other features constant, to project disease counts forward each year up to 2043.

Results

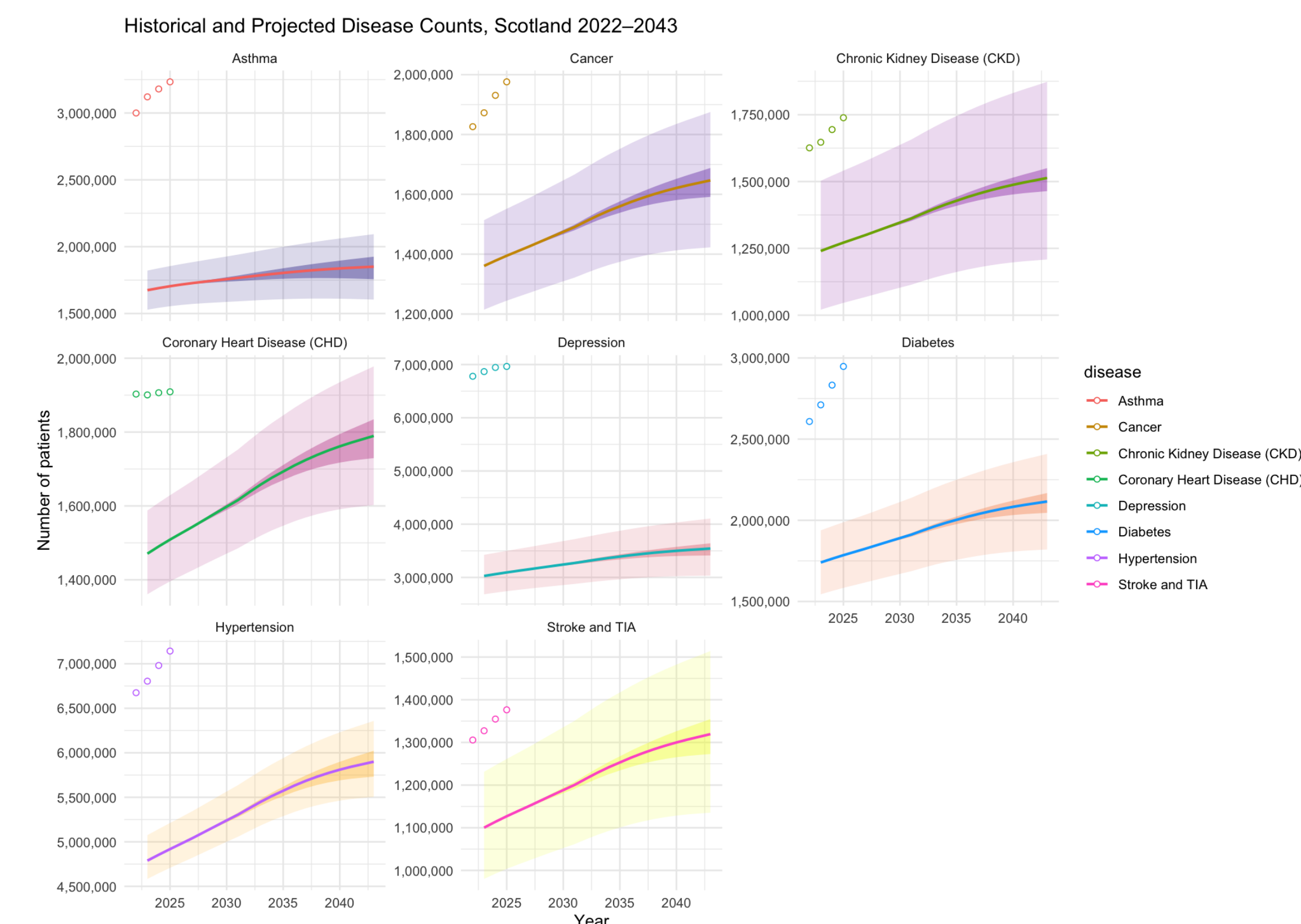


Figure 3. Historical and projected disease counts, Scotland 2022–2043. Points represent historical data; lines represent projections under the ONS principal scenario; inner ribbons show low–high population variants; outer ribbons show posterior predictive 95% credible intervals). The model underestimates relative to the historical trajectory.

Discussion

- **Demographic drivers.** Age and sex emerged as the strongest predictors, with cardiovascular and metabolic conditions increasing steeply in older cohorts, while depression remained higher in younger groups. These patterns confirm population ageing as the primary driver of future GP demand, rather than shifts in underlying risk.
- **Regional variation.** Projections highlight that boards and partnerships with older or more deprived populations will see disproportionate increases, suggesting resource allocation must be targeted rather than uniform. Cardiovascular disease, diabetes, and depression are likely to be central pressure points for GP services.
- **Limitations.** The model provides internally consistent scenarios rather than exact forecasts. Relative to observed prevalence, projections underestimated recent growth, likely due to short prevalence series, harmonised subnational scaling, and simplified assumptions about stable epidemiology. GP supply was proxied only by headcount, omitting working hours and skill mix, which may further bias capacity estimates.

References

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